Review Of EDA For Dataset vol_dataset_post_wrangle_021026.csv

Date: 2/11/2026

EDA Code Used: EDA_ETF_Volatility_TimeSeries_VB_021026.ipynb

EDA Input File:
- vol_dataset_post_wrangle_021026.csv

EDA Output Files:
- EDA_ETF_Volatility_TimeSeries_vb_done_021026_021026.ipynb
- EDA_ETF_Volatility_TimeSeries_vb_done_021026_021026_ipynb.pdf
- coverage_by_ticker_vb_021026.csv
- macro_constancy_check_vb_021026.csv
- missingness_by_column_vb_021026.csv

Platform Used To Run EDA Code:
- Google Colab

Description Of EDA Code In EDA_ETF_Volatility_TimeSeries_VB_021026.ipynb

This EDA notebook is for a time-series + panel ETF dataset (panel means ticker- and date-based). It's not "generic class EDA"; it's specifically aimed at:
- making sure the panel is valid (no duplicate keys, dates in order, no broken time gaps),
- checking time-series behavior (volatility clustering, regimes, autocorrelation),
- and doing leakage/alignment sanity checks so our target really uses "future" information and our features really use only "past/current" information.

<u>Inputs</u>

Primary input dataset:

Load one dataset (CSV or Parquet):
- DATA_PATH_CSV = "/content/drive/MyDrive/Colab Notebooks/vol_dataset_post_wrangle_021026.csv"
- or DATA_PATH_PARQUET = "... .parquet"
- USE_INPUT = "csv" or "parquet"

Expected key columns

- ticker, date (these define the panel key)

Core finance columns used in many checks/plots

- adj_close, volume, daily_log_return

Target (label)

- forward_vol_5d_annual_decimel_calculated (forward 5-day realized volatility target)

Features

- trailing vol (calculated + FactSet versions), and macro columns (US_3M_TB_YLD, VIX, etc.)

What the code does (section-by-section)

0) Imports

- Imports core libraries: pandas, numpy, matplotlib, seaborn and statsmodels.

1) Configuration + Drive mount

- Mounts Google Drive so /content/drive/... paths work.
- Centralizes all settings (paths, column names, sample ticker, cutoffs).
- Creates an output folder:
    - OUTPUT_DIR = "/content/drive/MyDrive/Colab Notebooks/outputs"
    - os.makedirs(OUTPUT_DIR, exist_ok=True)

What you should see if it worked: a message that Drive mounted, and the outputs directory exists.

2) Load dataset + parse dates + canonical sort

- Reads CSV (or Parquet).
- Converts date to true datetime:
- df[DATE_COL] = pd.to_datetime(..., errors="coerce")
- Asserts no bad date parses.
- Sorts by [ticker, date] ascending → oldest → newest within each ticker.

Why this matters: almost every time-series operation (shift, rolling) assumes correct chronological order. If reversed, you can silently create leakage.

What you should see if it worked:

- "Bad date parses: 0"
- A sensible date range (you had 2014-12-31 → 2025-12-31)

3) Panel integrity checks (keys, duplicates, monotone dates, coverage)

This is the “is my panel dataset sane?” block:

- Checks duplicate (ticker, date) rows:
  - df.duplicated(KEY_COLS).sum()
- Checks monotone increasing dates per ticker (no out-of-order dates).
- Builds coverage table per ticker: start date, end date, number of rows.

4) Gaps + missingness

Two separate concepts:

4A) Big gaps inside a ticker

- Computes day gaps between consecutive dates in each ticker.
- Flags gaps > BIG_GAP_DAYS (default 10 calendar days).

4B) Missingness

- Missing rate by column (df.isna().mean()).
- Missingness of the target by ticker (useful because labels often drop at the edges).

5) Macro “constancy” sanity check

- For each macro column, it computes nunique across tickers per date.
- Ideal:
  - median unique per date = 1
  - percent of dates with >1 unique value = 0

6) Distribution EDA (pooled histograms + summary stats)

Pools all tickers + all dates together and plots histograms + prints describe() tables for:

- daily log returns,
- log(volume),
- trailing vol columns,
- target forward vol.

This is mainly to understand:

- heavy tails,
- scale differences,
- and whether transformations are needed.

7) Single-ticker time-series plots (SPY by default)

For EDA_SAMPLE_TICKER = "SPY":

- price series,
- returns series,
- trailing vol series,
- target series,

- macro series overlays.

This is the "does it *look* like finance data?" check (vol clustering, regime changes, spikes around crisis periods).

8) Autocorrelation diagnostics (ACF-style)

For the sample ticker:
- ACF of returns (usually small),
- ACF of squared returns (often strong → volatility clustering),
- ACF of target vol (should show persistence).

Why it matters:

- In markets:
  - returns usually have low autocorrelation (close to white noise)
  - volatility (and squared returns) usually have positive autocorrelation (vol clustering / persistence)

How to interpret the plots:
- If returns ACF shows big spikes at many lags → suspicious (could be sorting/duplication/leakage).
- If squared returns / vol ACF shows slower decay → normal and desirable to confirm time-series structure.

If statsmodels exists, uses acf; otherwise uses a fallback correlation approach.

<u>Outputs (what files/plots are outputted)</u>

On-screen outputs inside the notebook
- Printed dataset shape, columns count, date parse status, date range.
- Printed duplicate counts, monotonicity results.
- Tables: coverage by ticker, missingness by column, target missingness by ticker, macro constancy table.
- Plots: histograms (returns/volume/vol), SPY time-series panels, ACF plots.

Saved-to-disk/drive outputs

The notebook is set up to save artifacts into:

- /content/drive/MyDrive/Colab Notebooks/outputs

CSV summaries include:
- coverage_by_ticker.csv
- missingness_by_column.csv

- macro_constancy_check.csv

<u>Leakage pitfalls this notebook is guarding against</u>

- Wrong date order (reversing dates breaks shift() / rolling() logic).
- Label misalignment (target accidentally using today's or past data, not future).
- Macro join issues (macro values varying by ticker on the same date indicates a bad merge).

EDA Results For vol_dataset_post_wrangle_021026.csv

1) Panel integrity checks (keys, duplicates, monotone dates, coverage)
   - Duplicate key rows: 0
   - on-monotone tickers: 0
   - Tickers: 36
   - Many tickers have 2767 rows (full history), some start later (for example BEDZ starts 2021).

2) Gaps
   - Big gaps: 0 (good).

3) Missingness
   - Macro columns showed 0 missing (good).
   - market missing very high.
   - sector / industry missing a lot.
   - Target missing was tiny (~0.2% overall).

4) Macro "constancy"
   - perfect constancy for all listed macro columns- good

#1 to #4 show that panel is structurally clean enough for time-series ML. Dataset passed the key failure modes (duplicate keys, broken sort order, bad joins, macro leakage).

5) Distribution EDA (pooled histograms + summary stats)
   - Daily log returns (pooled): mean ~0.00029, std ~0.016, with fat tails (min about -0.46, max about 0.198).
     This is "normal" for market data: returns are centered near 0 but have occasional big shocks.
   - Volume: extremely spread out (median ~1.33M, 99th percentile ~97.9M, max ~507M).
     This is exactly why log(volume) is plotted.

- Trailing vol + target forward vol: the histograms show "volatility clustering" (many small values with a long right tail).
- forward-vol target has rare big spikes (crisis-like days).

How to read those histograms:

- Tall peak near small vol + long right tail = volatility is usually calm, sometimes explodes.
- A giant spike at 0 for a volatility series, can indicate:
  - too-short rolling window at the start of each ticker (expected), or
  - a bad calculation / fill logic (worth checking).

6) Single-ticker time-series plots (SPY by default)

The notebook prints:

- SPY has 2767 rows, date range 2014-12-31 → 2025-12-31.
  That's a full daily-history-length series (good coverage).

The SPY plots show:

- rising price over time,
- big return spikes (notably around stress),
- volatility spikes (clustering),
- macro series trends (e.g., yields rising post-2021).

7) Autocorrelation diagnostics

ACF plots were run on SPY only.

Assume ACF conclusions are likely directionally correct across all ETFs in dataset.

ACF results show:

- Returns ACF ≈ 0 at most lags.
  Meaning: daily returns behave close to white noise (little linear predictability from past returns).
  Implication for project: don't expect past returns alone to forecast direction well.
- Absolute / squared returns show clear positive autocorrelation that decays slowly.
  Meaning: volatility clustering (big moves tend to follow big moves; calm follows calm).
  This is one of the core "stylized facts" of markets, and plots confirm dataset "looks like finance data."
  Implication: volatility is predictable in a statistical sense, even when returns are not.

- Forward 5-day realized volatility target is autocorrelated (persistent)
  Meaning: label (forward 5-day realized vol) has memory—high vol regimes persist across time.
  Implication: this is good news for modeling: there is real signal to learn.

Practical modeling takeaway.
EDA supports the thesis:

- Predict volatility, not return direction (at least as the first serious model).
- A strong baseline is persistence: "tomorrow's / next-week vol ≈ today's trailing vol."

ACF results justify that baseline and tell us it will be hard to beat without richer features.

8) On-screen outputs inside the notebook

- Printed dataset shape, columns count, date parse status, date range.
  - Loaded shape (96525, 23) → dataset size matches output expectation.
  - Bad date parses: 0 → date successfully converted to datetime.
  - Date range: 2014-12-31 → 2025-12-31 → global panel coverage.

- Printed duplicate counts, monotonicity results.
  - Duplicate (ticker,date) rows: 0 → our key is unique (critical for panel modeling).
  - Tickers with non-monotone dates: 0 → per-ticker ordering is clean (critical for rolling windows).
  - Big gaps > 10 days: 0 → no suspicious missing multi-week chunks inside tickers (good).

9) CSV summary coverage_by_ticker.csv

This output file answers "Do I have enough history per ETF?" by reporting:

- start_date, end_date: first / last date available for that ticker
- n_days: number of observations (trading days)

Our output shows:

- There are 36 tickers.
- Most tickers have a long history starting around 2014-12-31 through 2025-12-31.
- A handful start later (shorter history), for example:
  - BEDZ (starts 2022-03-02, ~959 rows)

    - XLC (starts 2018-06-19, ~1900 rows)
    - FTXG (starts 2016-09-15, ~2340 rows)
    - XLRE (starts 2015-10-09, ~2577 rows)
    - VGT (starts 2015-01-02, ~2760 rows)

10) CSV summary missingness_by_column.csv

This output file answers "Which columns are sparse and might break modeling?" by reporting:

- missing_rate (fraction missing)
- n_missing and n_total

Our output shows:

- market ~97% missing
- sector ~64% missing
- industry ~39% missing
- volume ~7.6% missing
- Everything else is basically ~0–0.1% missing (very clean).

11) CSV summary macro_constancy_check.csv

This output file answers ""Are macro values identical across tickers on the same date?"

This checks whether macro series (VIX, yields, oil, gold, etc.) are constant across tickers for a given date, which is what you want after joining macro features.

Our output shows for every macro column listed:

- median_unique_per_date = 1
- max_unique_per_date = 1
- pct_dates_unique_gt_1 = 0

This is perfect because on any given date, every ticker sees the same macro value.

Load dataset + parse dates + canonical sort

```
Loaded shape: (96525, 23)
Columns: 23
Bad date parses: 0
Date range: 2014-12-31 → 2025-12-31
```

	date	ticker	adj_close	has_price	volume_k	volume	trailing_vol_daily_pct_5d_factset	trailing_vol_daily_pct_20d_factset	trailing_vol_annual_decimel_5d_factset	trailing_vol_annual_decimel_20d_factset	...	group_type	market	sector	industry	daily_log_return	NYGOLDS	OIL_WTI_S	US_10Y_BOND_YLD	US_3M_TB_YLD	VIX
0	2021-04-21	BEDZ	24.95	1.0	84.476	84476.0	1.5753	1.5827	0.2501	0.2512	...	industry	NaN	NaN	hotels_resorts_cruise_lines	NaN	1793.6	61.34	1.56	0.03	17.50
1	2021-04-22	BEDZ	25.05	1.0	15.443	15443.0	1.5753	1.5827	0.2501	0.2512	...	industry	NaN	NaN	hotels_resorts_cruise_lines	0.004048	1783.9	61.45	1.56	0.03	18.71
2	2021-04-23	BEDZ	25.32	1.0	11.732	11732.0	1.5760	1.5833	0.2502	0.2513	...	industry	NaN	NaN	hotels_resorts_cruise_lines	0.010682	1776.9	62.18	1.56	0.02	17.33
3	2021-04-26	BEDZ	25.43	1.0	33.758	33758.0	1.5764	1.5837	0.2502	0.2514	...	industry	NaN	NaN	hotels_resorts_cruise_lines	0.004343	1781.0	62.02	1.57	0.02	17.64
4	2021-04-27	BEDZ	25.64	1.0	29.076	29076.0	1.5770	1.5843	0.2503	0.2515	...	industry	NaN	NaN	hotels_resorts_cruise_lines	0.008268	1775.8	63.03	1.62	0.01	17.56

5 rows × 23 columns

Panel integrity checks (keys, duplicates, monotone dates, coverage)

```
Duplicate (ticker,date) rows: 0
Tickers with non-monotone dates: 0
Number of tickers: 36
```

ticker	start	end	n_days
KBE	2014-12-31	2025-12-31	2767
IYZ	2014-12-31	2025-12-31	2767
KIE	2014-12-31	2025-12-31	2767
KCE	2014-12-31	2025-12-31	2767
XSD	2014-12-31	2025-12-31	2767
XME	2014-12-31	2025-12-31	2767
KRE	2014-12-31	2025-12-31	2767
PEJ	2014-12-31	2025-12-31	2767
PNQI	2014-12-31	2025-12-31	2767
PXQ	2014-12-31	2025-12-31	2767

ticker	start	end	n_days
XLV	2014-12-31	2025-12-31	2767
XLY	2014-12-31	2025-12-31	2767
XTN	2014-12-31	2025-12-31	2767
XPH	2014-12-31	2025-12-31	2767
XOP	2014-12-31	2025-12-31	2767
VGT	2015-01-02	2025-12-31	2766
XLRE	2015-10-08	2025-12-31	2573
FTXG	2016-09-21	2025-12-31	2333
XLC	2018-06-19	2025-12-31	1895
BEDZ	2021-04-21	2025-12-31	1181

Gaps and missingness (time continuity + feature missing rates)

```
Big gaps (gap_days > 10 ): 0
```

ticker	date	prev_date	gap_days

	missing_rate	missing_count
NYGOLDS	0.000000	0
OIL_WTI_S	0.000000	0
US_10Y_BOND_YLD	0.000000	0
US_3M_TB_YLD	0.000000	0
VIX	0.000000	0
adj_close	0.000000	0
daily_log_return	0.000373	36
date	0.000000	0
forward_vol_5d_annual_decimel_calculated	0.001865	180
group_type	0.000000	0
has_price	0.000000	0
industry	0.390272	37671
market	0.971334	93758
sector	0.638394	61621
ticker	0.000000	0

ticker	forward_vol_5d_annual_decimel_calculated
BEDZ	0.004234
XLC	0.002639
FTXG	0.002143
XLRE	0.001943
VGT	0.001808
IYZ	0.001807
KBE	0.001807
KRE	0.001807
PEJ	0.001807
PNQI	0.001807

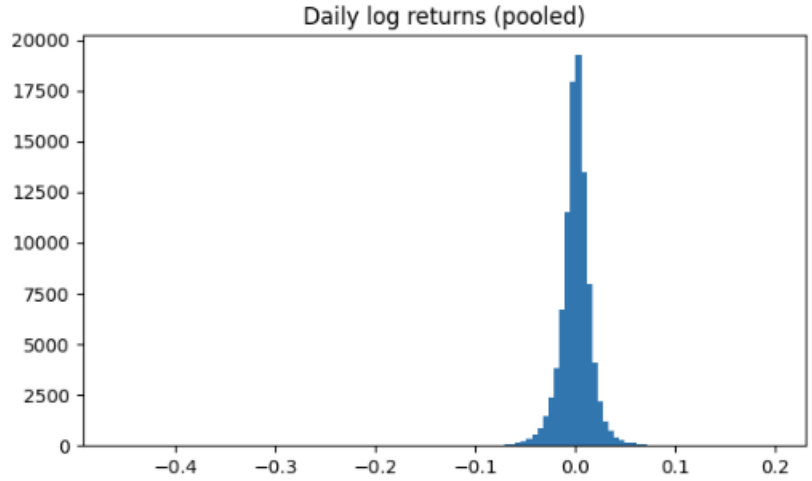
dtype: float64

Macro feature sanity (should be constant across tickers on a given date)

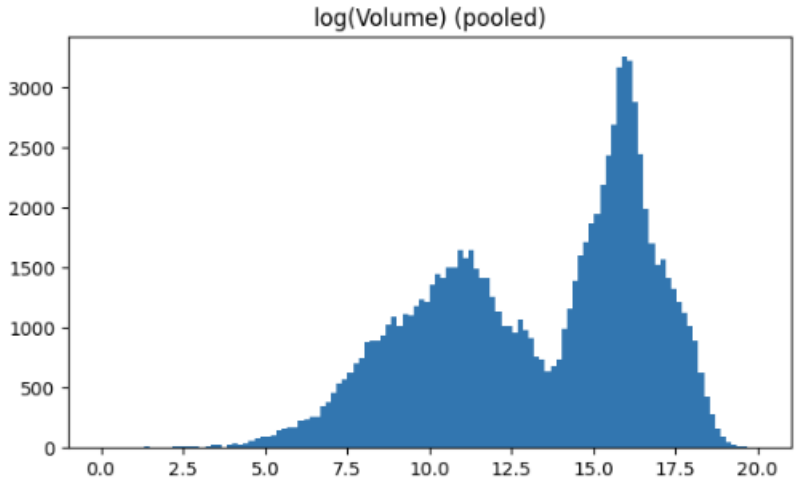
```
Macro columns present: ['US_3M_TB_YLD', 'US_10Y_BOND_YLD', 'NYGOLDS', 'OIL_WTI_S', 'VIX']
```

	median_unique_per_date	max_unique_per_date	pct_dates_unique_gt_1
US_3M_TB_YLD	1.0	1.0	0.0
US_10Y_BOND_YLD	1.0	1.0	0.0
NYGOLDS	1.0	1.0	0.0
OIL_WTI_S	1.0	1.0	0.0
VIX	1.0	1.0	0.0

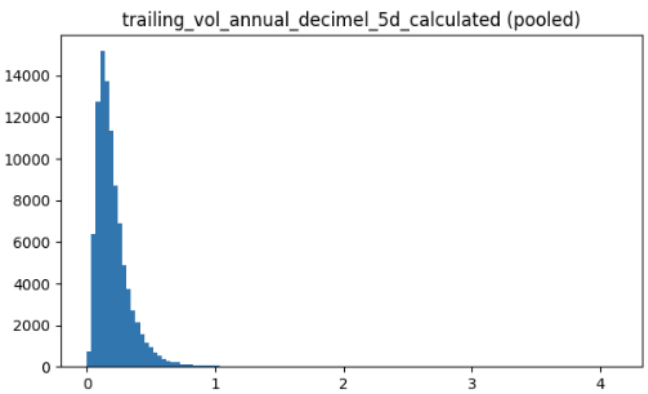
Distribution EDA (returns, volume, trailing vol, target vol)



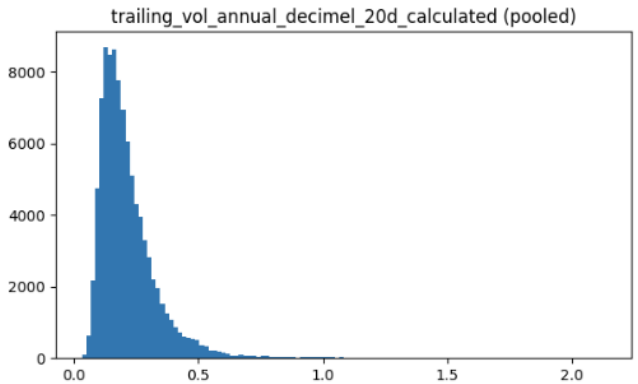

```
count    96489.000000
mean         0.000290
std          0.016145
min         -0.459984
1%          -0.044887
5%          -0.024899
50%          0.000772
95%          0.023346
99%          0.042107
max          0.197543
Name: daily_log_return, dtype: float64
```



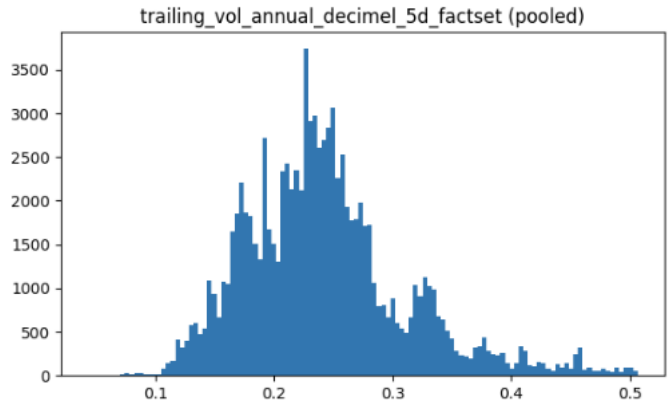

```
count    9.597800e+04
mean     9.397789e+06
std      2.049194e+07
min      1.000000e+00
1%       2.690000e+02
5%       1.915000e+03
50%      1.328570e+06
95%      4.827411e+07
99%      9.794572e+07
max      5.072443e+08
Name: volume, dtype: float64
```



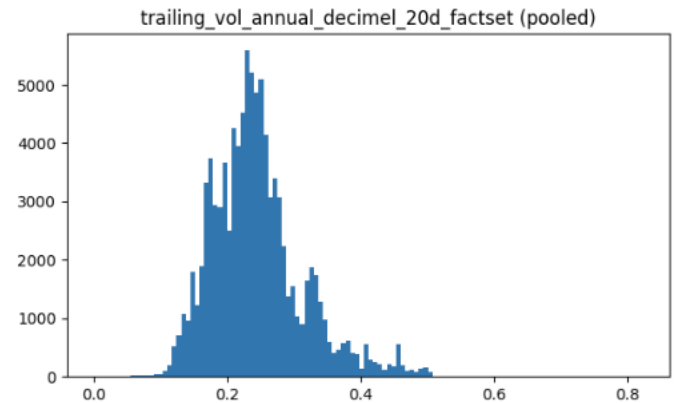

```
trailing_vol_annual_decimel_5d_calculated {'count': 96345.0, 'mean': 0.2069985458508485, 'std': 0.15347008081123947, 'min': 0.0, '1%': 0.0371, '5%': 0.0601, '50%': 0.1701, '95%': 0.4669, '99%': 0.7661039999999979, 'max': 4.1207}
```



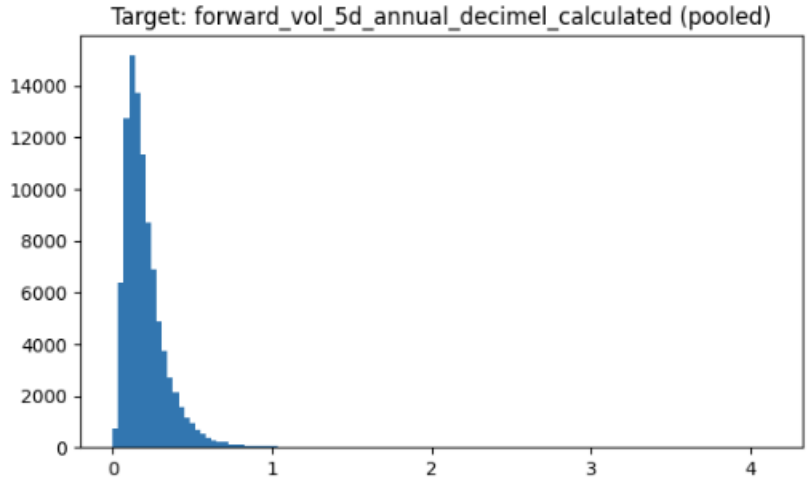
trailing_vol_annual_decimel_20d_calculated {'count': 95805.0, 'mean': 0.2201050028704138, 'std': 0.1324635772389395, 'min': 0.0319, '1%': 0.0699, '5%': 0.0931, '50%': 0.1882, '95%': 0.4478, '99%': 0.7391800000000032, 'max': 2.1347}



trailing_vol_annual_decimel_5d_factset {'count': 96525.0, 'mean': 0.24293003367003368, 'std': 0.06871785874580691, 'min': 0.0435, '1%': 0.1203, '5%': 0.1472, '50%': 0.2351, '95%': 0.3761, '99%': 0.4582, 'max': 0.5063}



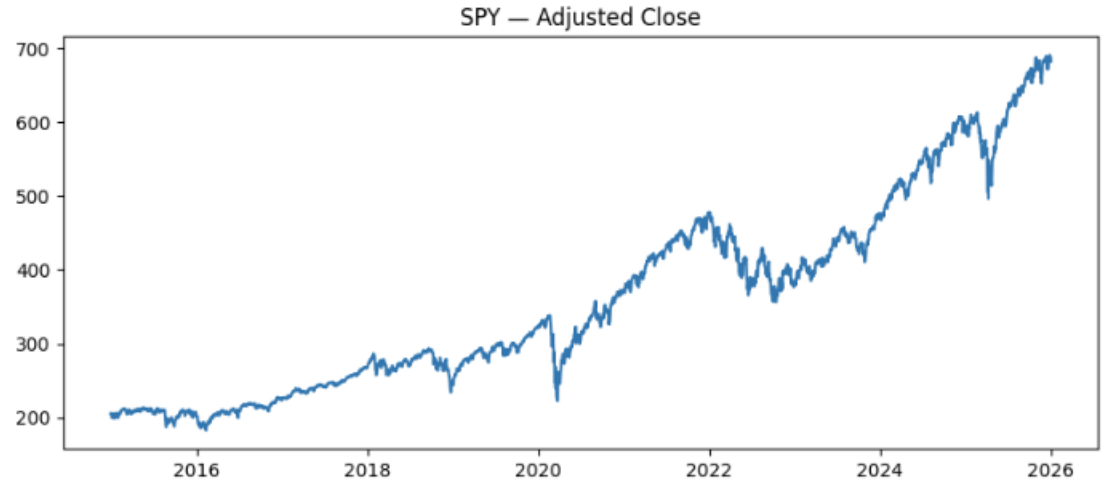
trailing_vol_annual_decimel_20d_factset {'count': 96489.0, 'mean': 0.24402550549803598, 'std': 0.06884942270723506, 'min': 0.0001, '1%': 0.1215, '5%': 0.1481, '50%': 0.2359, '95%': 0.3774, '99%': 0.4589, 'max': 0.8234}

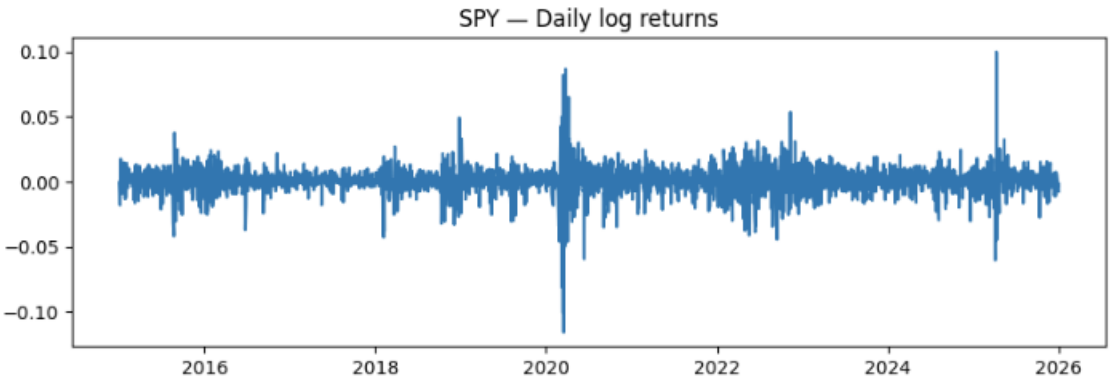


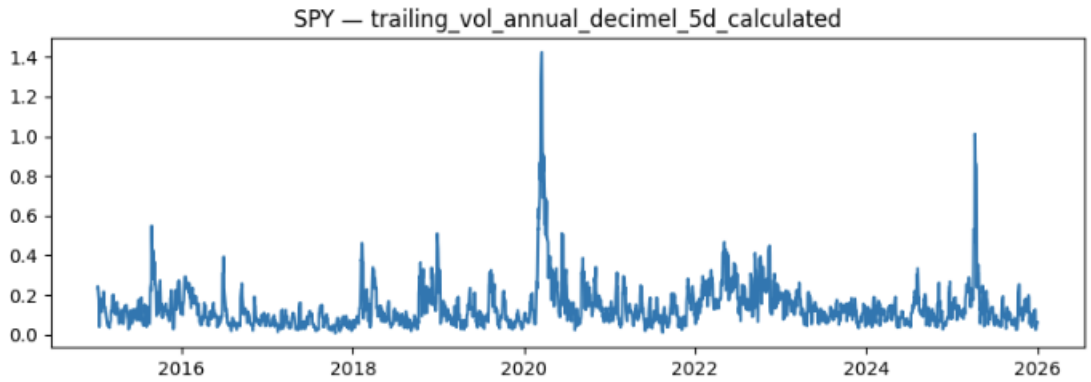

```
count    96345.000000
mean         0.206999
std          0.153470
min          0.000000
1%           0.037100
5%           0.060100
50%          0.170100
95%          0.466900
99%          0.766104
max          4.120700
Name: forward_vol_5d_annual_decimel_calculated, dtype: float64
```

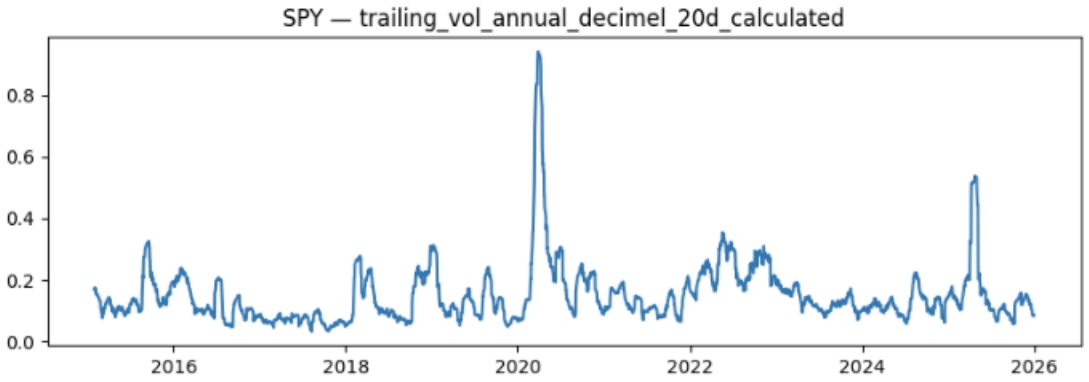
Time-series plots for one ticker (price, returns, trailing vol, forward vol)

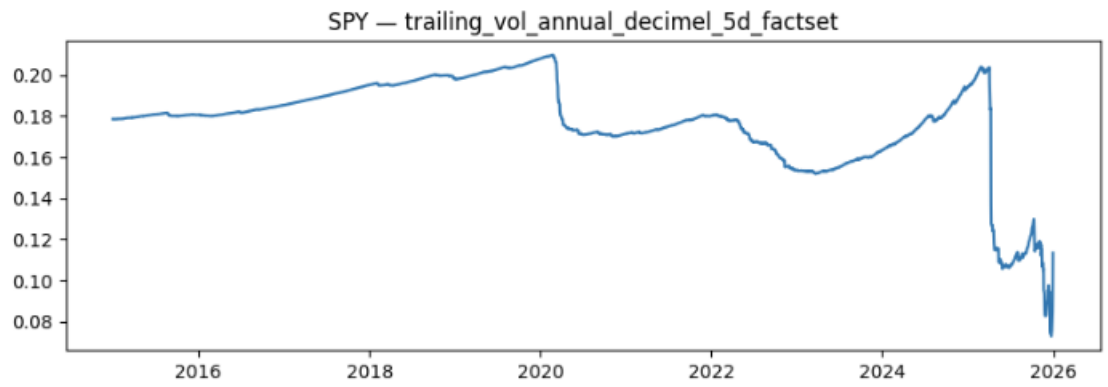
```
Ticker: SPY | rows: 2767 | range: 2014-12-31 → 2025-12-31
```

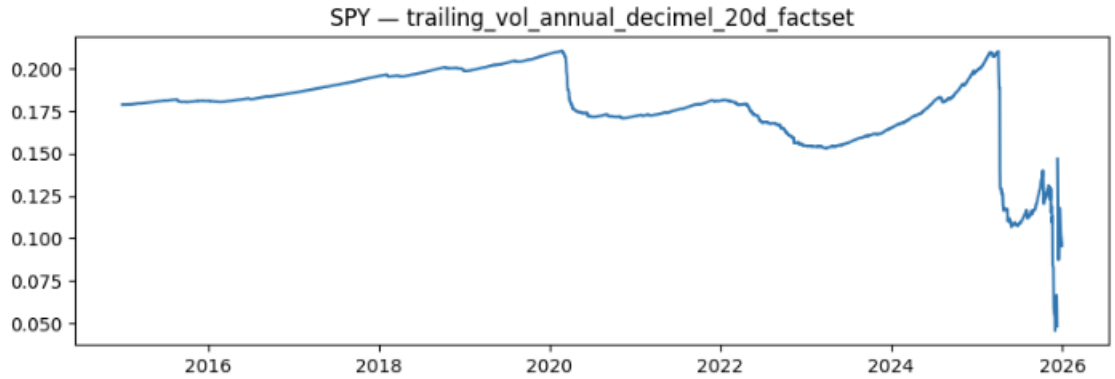


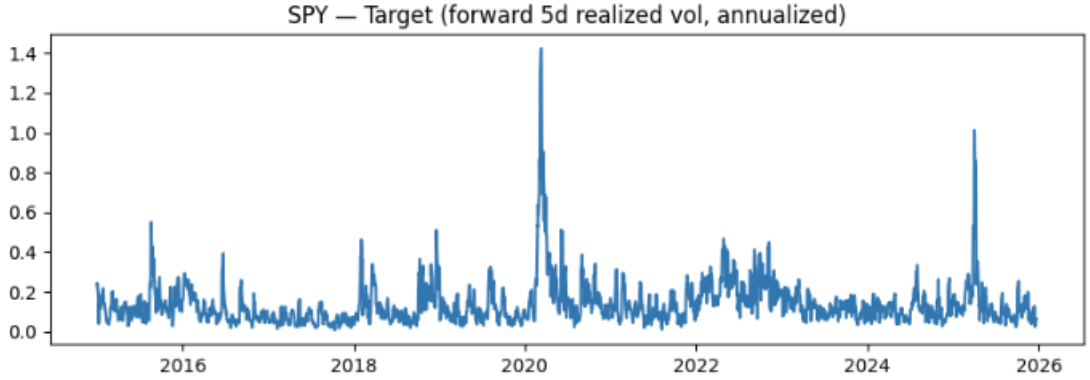


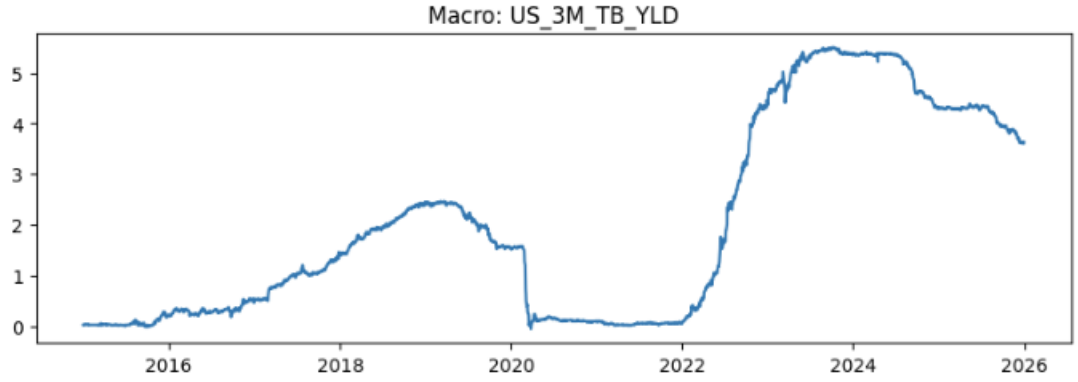

SPY — trailing_vol_annual_decimel_5d_calculated
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2016
2018
2020
2022
2024
2026

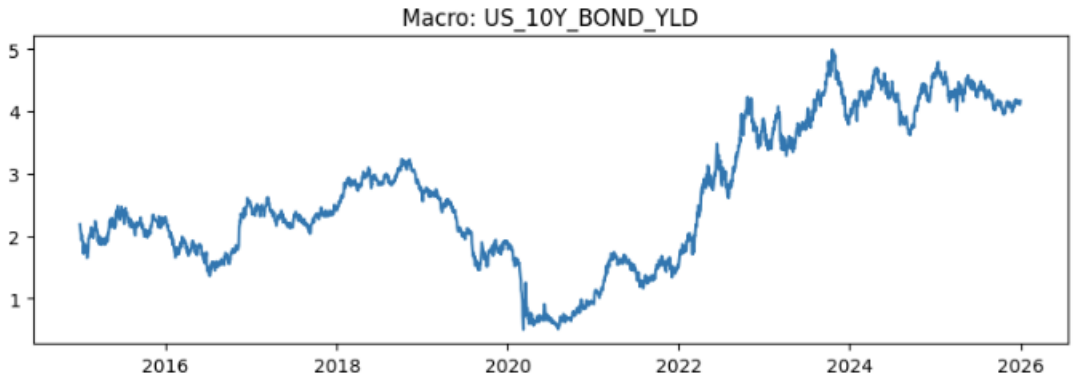

SPY — trailing_vol_annual_decimel_20d_calculated
0.8
0.6
0.4
0.2
0.0
2016
2018
2020
2022
2024
2026

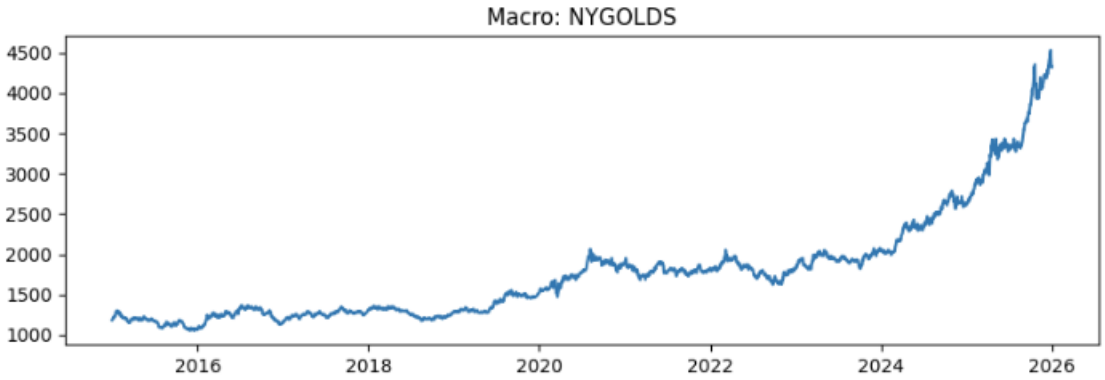

SPY — trailing_vol_annual_decimel_5d_factset
0.20
0.18
0.16
0.14
0.12
0.10
0.08
2016
2018
2020
2022
2024
2026

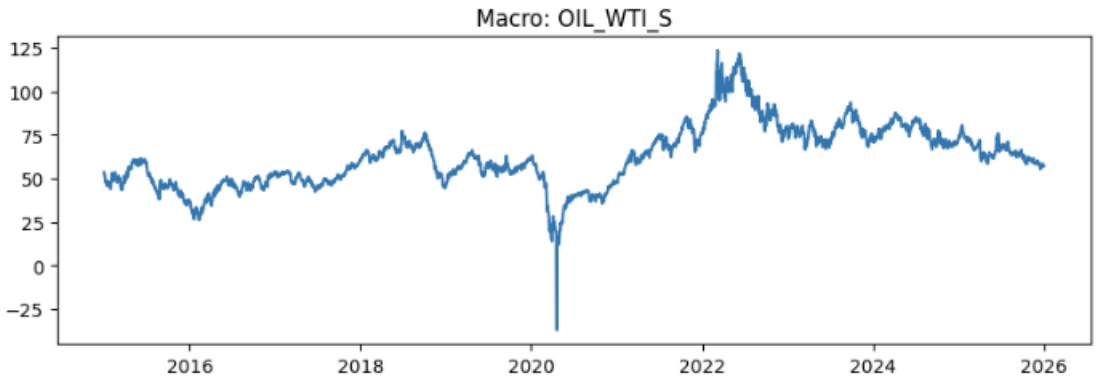

SPY — trailing_vol_annual_decimel_20d_factset
0.200
0.175
0.150
0.125
0.100
0.075
0.050
2016
2018
2020
2022
2024
2026

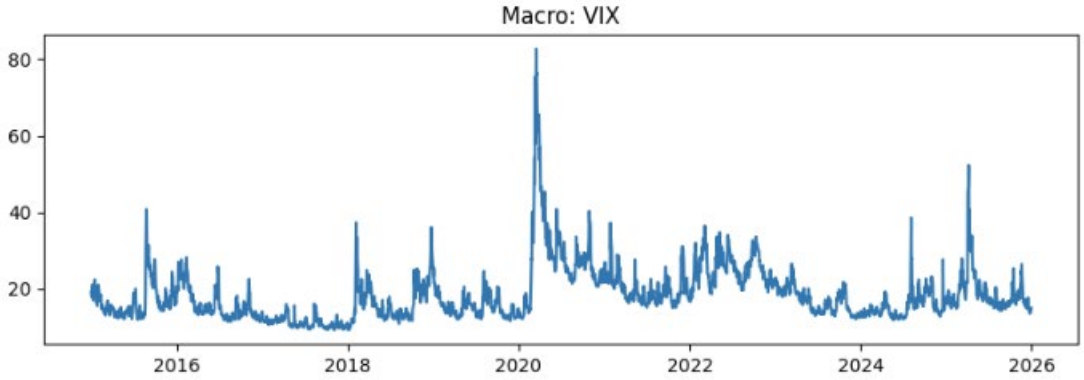

SPY — Target (forward 5d realized vol, annualized)
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2016
2018
2020
2022
2024
2026


Macro: US_3M_TB_YLD
5
4
3
2
1
0
2016
2018
2020
2022
2024
2026

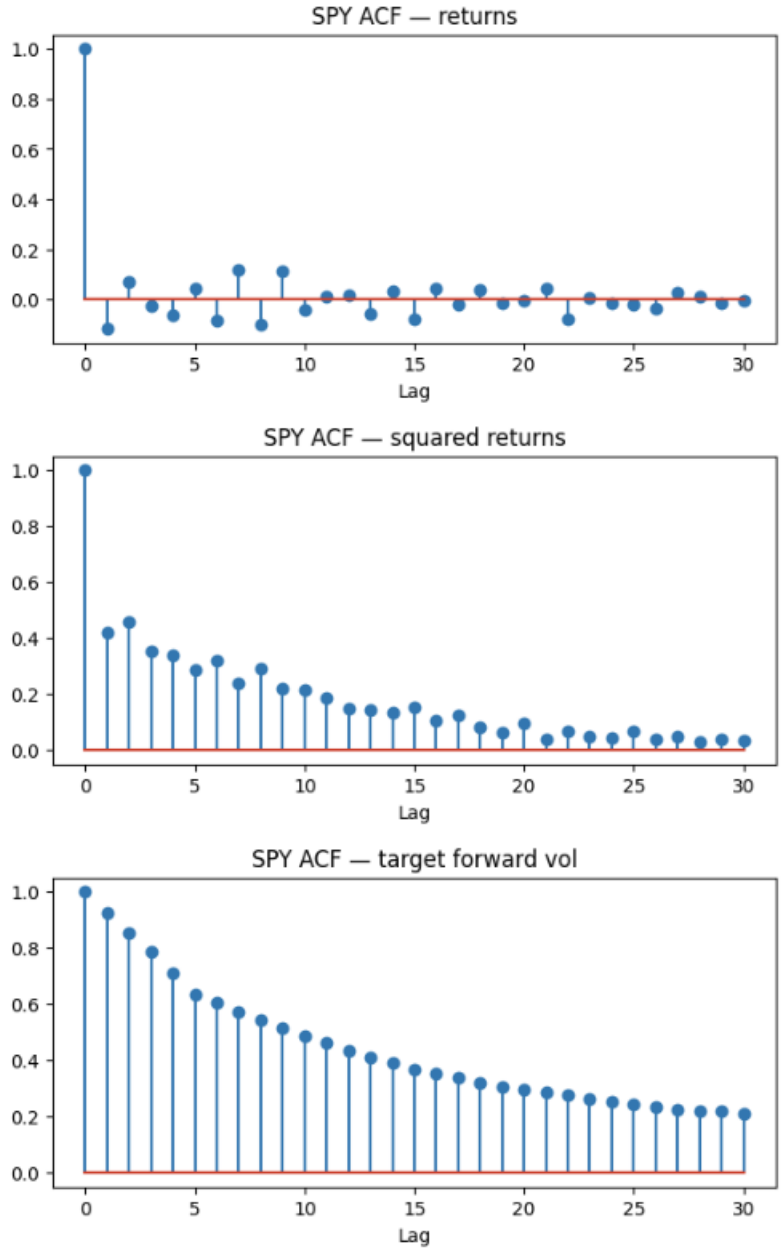

Macro: US_10Y_BOND_YLD
5
4
3
2
1
2016
2018
2020
2022
2024
2026


Macro: NYGOLDS
4500
4000
3500
3000
2500
2000
1500
1000
2016
2018
2020
2022
2024
2026


Macro: OIL_WTI_S
125
100
75
50
25
0
−25
2016
2018
2020
2022
2024
2026


Macro: VIX
80
60
40
20
2016
2018
2020
2022
2024
2026

Autocorrelation diagnostics (returns vs squared returns vs volatility)

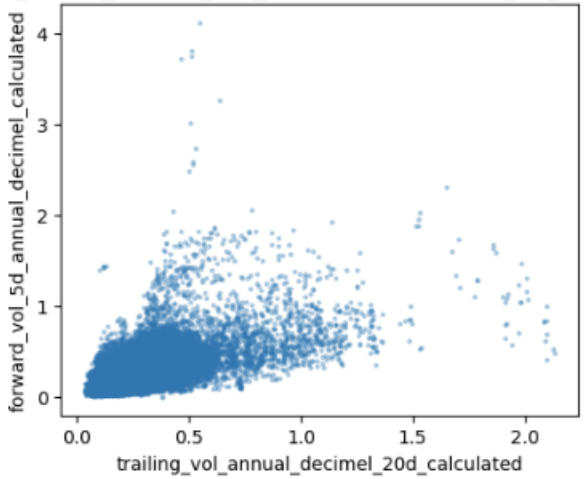


Feature–target relationships (scatter + correlations)

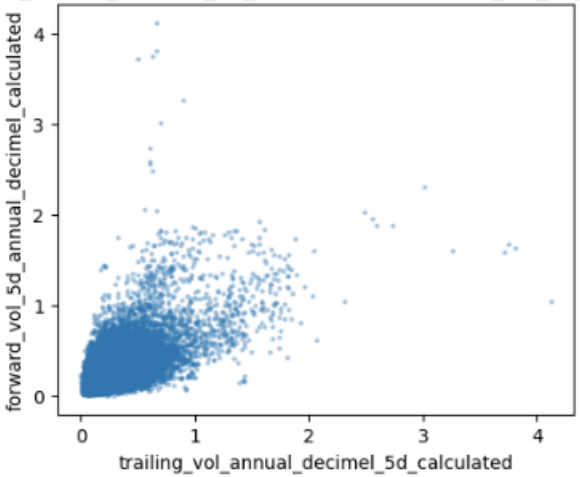
```
Train rows: 69453 | Train end: 2022-12-31
Candidate feature columns: ['trailing_vol_annual_decimel_5d_calculated', 'trailing_vol_annual_decimel_20d_calculated', 'trailing_vol_annual_decimel_5d_factset', 'trailing_vol_annual_decimel_20d_factset', 'US_3M_TB_YLD', 'US_10Y_BOND_YLD', 'NYGOLDS', 'OIL_WTI_S', 'VIX']
```

	forward_vol_5d_annual_decimel_calculated
forward_vol_5d_annual_decimel_calculated	1.000000
trailing_vol_annual_decimel_20d_calculated	0.642738
trailing_vol_annual_decimel_5d_calculated	0.636462
VIX	0.591595
trailing_vol_annual_decimel_20d_factset	0.330381
trailing_vol_annual_decimel_5d_factset	0.329740
NYGOLDS	0.227679
OIL_WTI_S	-0.017132
US_3M_TB_YLD	-0.041820
US_10Y_BOND_YLD	-0.128753

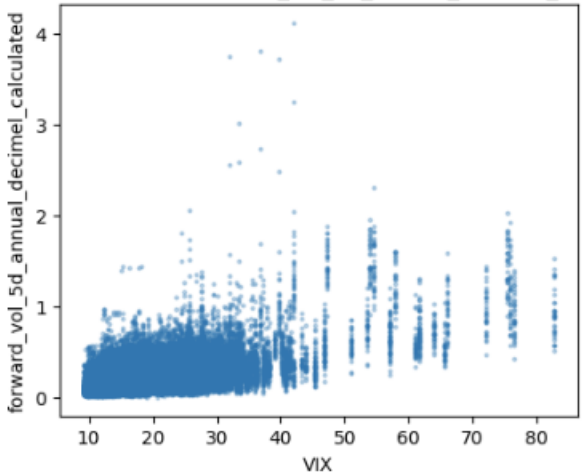
Train scatter: trailing_vol_annual_decimel_20d_calculated vs forward_vol_5d_annual_decimel_calculated



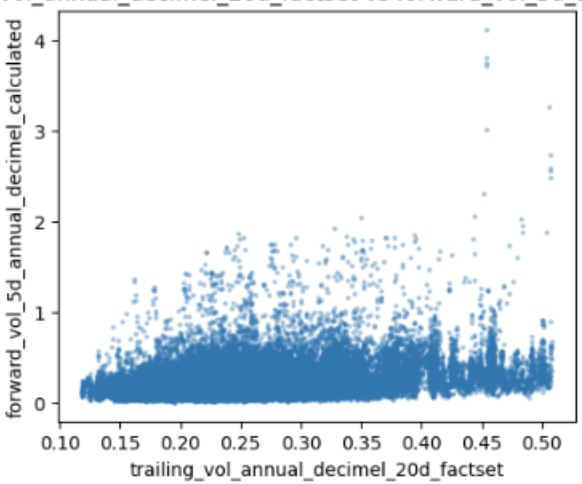
Train scatter: trailing_vol_annual_decimel_5d_calculated vs forward_vol_5d_annual_decimel_calculated



Train scatter: VIX vs forward_vol_5d_annual_decimel_calculated

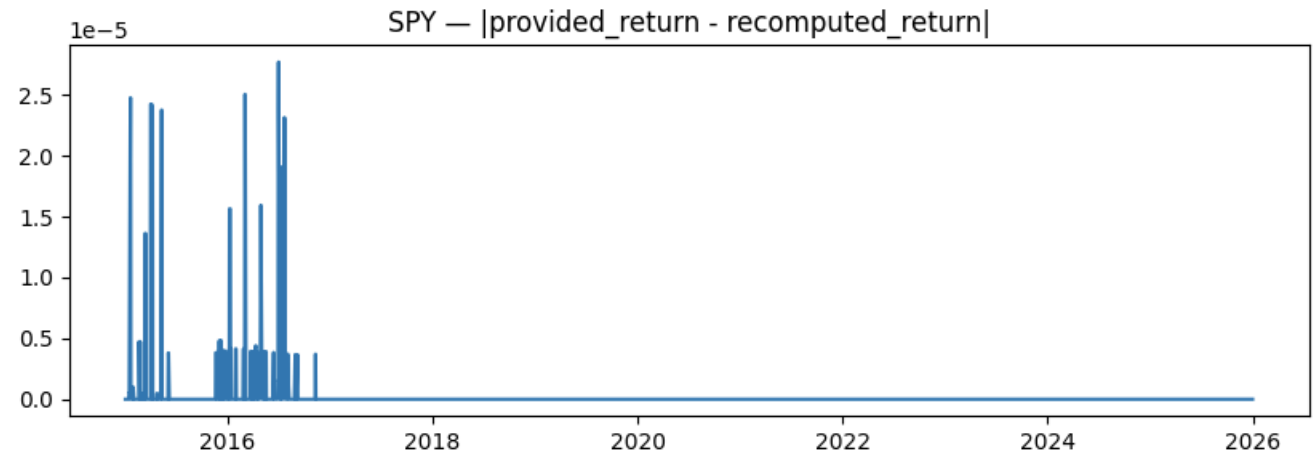


Train scatter: trailing_vol_annual_decimel_20d_factset vs forward_vol_5d_annual_decimel_calculated

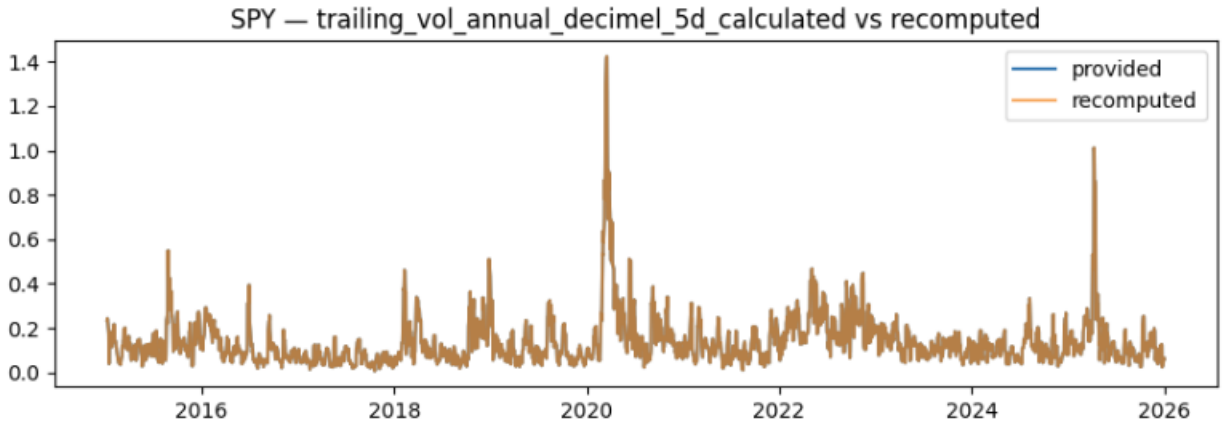


Alignment & leakage sanity checks (recompute returns, trailing vol, forward vol)

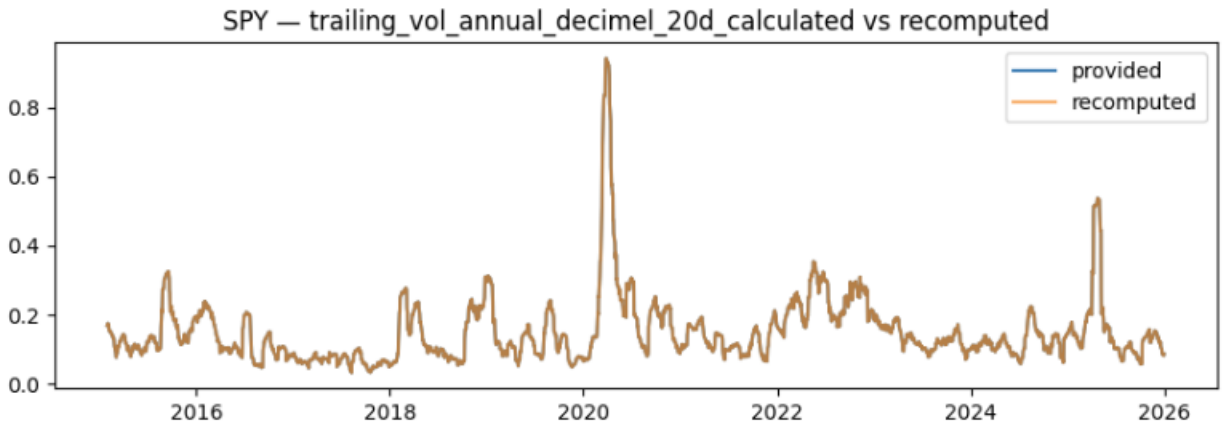
```
Return check — mean abs diff: 2.5330740337401045e-07 | median abs diff: 2.54220554329887e-10
```

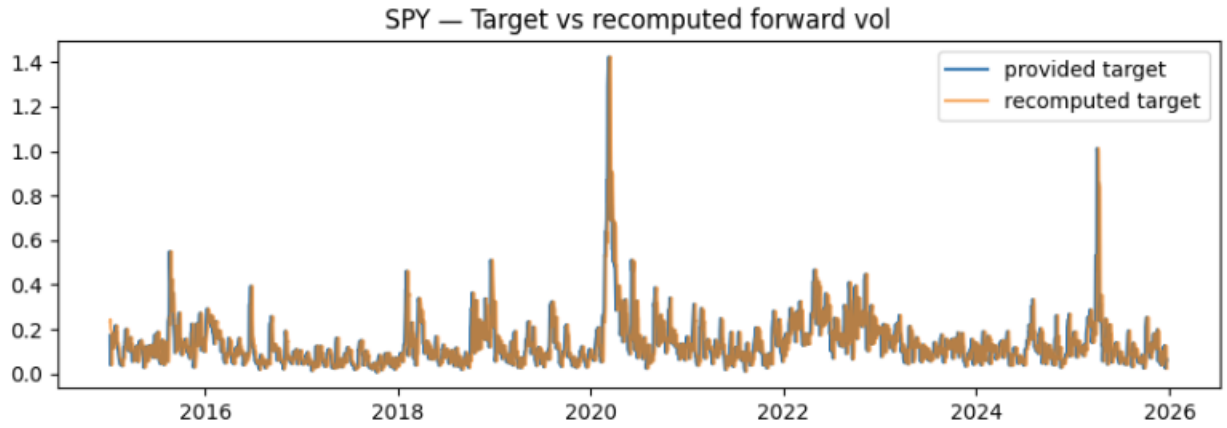


```
DEBUG: Columns in 'valid' for trailing_vol_annual_decimel_5d_calculated: ['date', 'trailing_vol_annual_decimel_5d_calculated', 'trail_vol_5d_recomp']
SPY trailing vol check — corr(trailing_vol_annual_decimel_5d_calculated, trail_vol_5d_recomp) = 1.0
```




```
DEBUG: Columns in 'valid' for trailing_vol_annual_decimel_20d_calculated: ['date', 'trailing_vol_annual_decimel_20d_calculated', 'trail_vol_20d_recomp']
SPY trailing vol check — corr(trailing_vol_annual_decimel_20d_calculated, trail_vol_20d_recomp) = 1.0
```




SPY — Target vs recomputed forward vol
provided target
recomputed target
1.4
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2016
2018
2020
2022
2024
2026

Simple baselines to beat (persistence using trailing volatility)

```
Baseline candidate predictors: ['trailing_vol_annual_decimel_5d_calculated', 'trailing_vol_annual_decimel_20d_calculated']
```

	baseline_feature	split	n	RMSE	MAE
5	trailing_vol_annual_decimel_20d_calculated	test	8820	0.154276	0.097534
3	trailing_vol_annual_decimel_20d_calculated	train	68733	0.132880	0.085128
4	trailing_vol_annual_decimel_20d_calculated	val	18072	0.092954	0.068025
2	trailing_vol_annual_decimel_5d_calculated	test	8820	0.161874	0.102006
0	trailing_vol_annual_decimel_5d_calculated	train	69273	0.140559	0.093647
1	trailing_vol_annual_decimel_5d_calculated	val	18072	0.109418	0.080194

coverage_by_ticker_vb_021126.csv

	A	B	C	D
1	ticker	start	end	n_days
2	KBE	12/31/2014	12/31/2025	2767
3	IYZ	12/31/2014	12/31/2025	2767
4	KIE	12/31/2014	12/31/2025	2767
5	KCE	12/31/2014	12/31/2025	2767
6	XSD	12/31/2014	12/31/2025	2767
7	XME	12/31/2014	12/31/2025	2767
8	KRE	12/31/2014	12/31/2025	2767
9	PEJ	12/31/2014	12/31/2025	2767
10	PNQI	12/31/2014	12/31/2025	2767
11	PXQ	12/31/2014	12/31/2025	2767
12	SPY	12/31/2014	12/31/2025	2767
13	XAR	12/31/2014	12/31/2025	2767
14	XHE	12/31/2014	12/31/2025	2767
15	XBI	12/31/2014	12/31/2025	2767
16	XES	12/31/2014	12/31/2025	2767
17	XHB	12/31/2014	12/31/2025	2767
18	XLB	12/31/2014	12/31/2025	2767
19	XHS	12/31/2014	12/31/2025	2767
20	XLF	12/31/2014	12/31/2025	2767
21	XLE	12/31/2014	12/31/2025	2767
22	XTL	12/31/2014	12/31/2025	2767
23	XSW	12/31/2014	12/31/2025	2767
24	XLI	12/31/2014	12/31/2025	2767
25	XLK	12/31/2014	12/31/2025	2767
26	XLP	12/31/2014	12/31/2025	2767
27	XLU	12/31/2014	12/31/2025	2767
28	XLV	12/31/2014	12/31/2025	2767
29	XLY	12/31/2014	12/31/2025	2767
30	XTN	12/31/2014	12/31/2025	2767
31	XPH	12/31/2014	12/31/2025	2767
32	XOP	12/31/2014	12/31/2025	2767
33	VGT	1/2/2015	12/31/2025	2766
34	XLRE	10/8/2015	12/31/2025	2573
35	FTXG	9/21/2016	12/31/2025	2333
36	XLC	6/19/2018	12/31/2025	1895
37	BEDZ	4/21/2021	12/31/2025	1181

missingness_by_column_vb_021126.csv

	A	B	C
1		missing_rate	missing_count
2	NYGOLDS	0	0
3	OIL_WTI_S	0	0
4	US_10Y_BOND_YLD	0	0
5	US_3M_TB_YLD	0	0
6	VIX	0	0
7	adj_close	0	0
8	daily_log_return	0.00037296	36
9	date	0	0
10	forward_vol_5d_annual_decimel_calculated	0.001864802	180
11	group_type	0	0
12	has_price	0	0
13	industry	0.39027195	37671
14	market	0.971333851	93758
15	sector	0.638394198	61621
16	ticker	0	0
17	trailing_vol_annual_decimel_20d_calculated	0.007459207	720
18	trailing_vol_annual_decimel_20d_factset	0.00037296	36
19	trailing_vol_annual_decimel_5d_calculated	0.001864802	180
20	trailing_vol_annual_decimel_5d_factset	0	0
21	trailing_vol_daily_pct_20d_factset	0.00037296	36
22	trailing_vol_daily_pct_5d_factset	0	0
23	volume	0	0
24	volume_k	0	0

macro_constancy_check_vb_021126.csv

	A	B	C	D
1		median_unique_per_date	max_unique_per_date	pct_dates_unique_gt_1
2	US_3M_TB_YLD	1	1	0
3	US_10Y_BOND_YLD	1	1	0
4	NYGOLDS	1	1	0
5	OIL_WTI_S	1	1	0
6	VIX	1	1	0